

## SensorMultiHeadDecoder Architecture (20260511)

Per-Row G-Code Recovery on Frozen Multi-Modal Sensor Encoder · ~39.9M total parameters (31.28M decoder + 8.6M frozen encoder)

## 64-second Sensor Window (T=256 samples @ 4 Hz, C=98 channels across 7 modality groups)

accel (18) gyro (18) magneto (18) temp (6) color (24) audio-RMS (6) motor-current (8) — from 6 distributed sensor units (frame\_l2/l3/r2, spindle2, y\_bed\_\_3/4)  
shape: (B, 256, 98)

## MM-DTAE-LSTM Encoder (8.6M params, FROZEN)

prior contribution; not retrained in this paper

## Per-Modality MLPs

7 projections → d=256  
LN + GELU + Dropout  
+ learned modality gates

## Cross-Modal MHA

multi-head self-attention  
denoising objective  
σ-Gaussian input noise

## 2-Layer LSTM

hidden=256  
temporal accumulation  
outputs memory tensor

## FROZEN

96.3% 9-class  
operation-class  
accuracy (prior work)Encoder Memory  $M \in \mathbb{R}^{256 \times 256}$ 

## Transformer Decoder (31.28M params)

8 layers, d\_model=384, 12 heads/layer, GELU FFN, dropout 0.1, pre-LayerNorm

Decoder Layer  $\ell$  (× 8 stacked)

## Masked Self-Attention

causal mask over  $y_{\{1:t-1\}}$ 

## Cross-Attention

Q=hidden, K,V=memory M

## Feed-Forward (GELU)

d\_ff=1536, dropout 0.1

residual + LayerNorm after each sublayer (pre-norm)

## Token Embedding + Positional Encoding

vocab=2,418 · learned sinusoidal positions

## Scheduled Sampling (training)

p\_max swept  $\in \{0, .25, .5, .75\}$ ; SS=0 deployedDecoder Hidden State  $h_t \in \mathbb{R}^{384}$ 

shared across all six heads

## Token Head

PRIMARY · inference

2,418-class softmax  
over full vocabulary  
bigram + FSM mask applied

## Type Head

auxiliary, multi-task

4-way: SPECIAL / CMD /  
PARAM / NUMERIC  
per-position classifier

## Motion-Command

auxiliary, multi-task

6-way: G0 / G1 / G2 /  
G3 / G53 / M30  
per-COMMAND position

## Parameter-Type

auxiliary, multi-task

10-way: X / Y / Z /  
I / J / K / R / F / S / P  
per-PARAM position

## Sign Head

auxiliary, multi-task

3-way:  
POS / NEG / ABSENT  
per-axis-value

## Per-Position Digit Head

auxiliary, regression-style

6 positions × 11 classes  
(digits 0–9 + PAD)  
complements bucket softmax

## Grammar-Constrained Decoding (inference time only)

Bigram Mask  $M_b \in \{0,1\}^{|V| \times |V|}$ first-order ordering (CMD→PARAM→NUM...)  
|V|=2,418 → avg 31.8 valid next tokens

## FSM Mask (state-dependent)

G0/G1 forbid R/I/J; M30 forbids continuation  
tracks last move-command as state

## Apply to Token-Head logits before softmax

 $\ell_t \leftarrow \ell_t \cdot M_b[y_{t-1}, :] \cdot \text{FSM\_mask}(s_t)$   
100% V1-violation elimination at no accuracy costG-Code Token Sequence Output  $\hat{y} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_L)$ 

BOS G1 X NUM\_X\_1492 Y NUM\_Y\_1485 F NUM\_F\_22 EOS

per-row target: one G-code line per sensor window; ~5–15 tokens typical

## Training Configuration

**Optimizer:** AdamW ( $\beta = 0.9$ ,  $\beta = 0.999$ ), lr= $5 \times 10^{-5}$ , weight decay 0.05

**Schedule:** 300 epochs, 10-epoch linear warm-up, early stop on val-token-acc (patience 75)

**Batch size:** 64 (full-window) or 4 (per-row long sequences); grad-clip norm 1.0

**Loss:** multi-head cross-entropy; default  $\lambda_{\text{tok}}=3.0$ , all auxiliary  $\lambda=1.0$ ; digit-head dw swept  $\in \{0, 1.0\}$

**CV:** 5-fold file-level stratified, coverage-repaired (97.6%/99.0% verbatim test-line overlap)

**Deployed config:** K=2,418 vocab, SS=0/dw=0 (T3.1 matched-methodology winner; +15.2 pp AR token vs SS=0.5 headline)

## Parameter Accounting

**Frozen encoder:** 8.6M params (MM-DTAE-LSTM, prior work; trained on 9-class operation classification)

**Decoder Transformer stack:** 18.9M params (8 layers × ≈ 2.4M per layer at d=384, 12 heads)

**Heads + embeddings + grammar:** 12.4M params (token-vocab projection 0.9M + auxiliary heads + token / positional embeddings)

total trainable in this paper: 31.28M / total system (frozen + trainable): ≈ 39.9M

**Legend:**   frozen module (no gradient)   trainable decoder block   primary inference head   auxiliary multi-task head   inference-only operation